

ABHIMITHRA PEDDI

AI / Machine Learning Engineer — Generative AI, LLMs & Agentic Systems

Hyderabad, India | +91 96034 46511 | abhimithrapeddi07@gmail.com

linkedin.com/in/abhi-mithra-peddi624b64158 | github.com/abhimithra02

PROFESSIONAL SUMMARY

AI/ML Engineer with 6+ years of professional experience spanning HR operations, data science, and production-grade Machine Learning & Generative AI systems. Currently designing agentic AI workflows where LLM-driven agents invoke tools via function calling to automate website QA at scale, alongside a RAG document intelligence system (LangChain, Pinecone, hybrid Vector + BM25 retrieval) that reduced manual document search time by 80%+. Owns both the modelling and the infrastructure: SageMaker retraining pipelines, Lambda/EC2/S3/RDS, IAM & encryption, Flask REST services, and structured run logging that makes every automated execution traceable and auditable.

TECHNICAL SKILLS

- **Generative AI & LLMs:** LLM Application Development, Prompt Engineering & Strict Output Parsing, Embeddings, RAG, Chunking Strategies, Hybrid Search (Vector + BM25), Token Management, Dynamic Prompt Truncation, Response Caching, LangChain
- **Agentic AI:** AI Agents & Agentic Workflows, Tool/Function Calling, Multi-Step Task Orchestration, Rule-Based Fallback Logic, Autonomous Validation Pipelines
- **ML & Deep Learning:** PyTorch, TensorFlow, Keras, Scikit-learn, CNNs, RNNs, NLP, Computer Vision (OpenCV, MediaPipe), Time-Series Forecasting, Predictive Modelling, Feature Engineering, Hyperparameter Tuning
- **Cloud & Deployment (AWS):** SageMaker, EC2, Lambda, S3, RDS, IAM, Lex, Rekognition, Polly, Encryption & Access Control, Docker, CI/CD
- **APIs & Backend:** Python (Advanced), Flask, Django, REST API Design, Streamlit, React.js
- **Data & Databases:** SQL, MySQL, MongoDB, Pinecone (Vector DB), Pandas, NumPy
- **Dev Tools:** Git, GitHub Copilot, ChatGPT/Claude, Jupyter, VS Code, PyCharm

PROFESSIONAL EXPERIENCE

Machine Learning Engineer | *vSplash Techlabs Private Limited, Hyderabad* Sep 2023 – Sep 2026

- Built an Agentic AI-powered QA automation platform for website testing where LLM-driven agents invoke validation tools through function calling with deterministic rule-based fallback, eliminating page-by-page manual review across client sites.
- Designed a meta-model validation layer that scores overall site quality and flags issues for human review, substantially reducing back-and-forth in QA cycles.
- Engineered advanced prompt templates with strict output parsing, ensuring the LLM reliably returns correctly structured SEO metadata and content schemas — removing hallucinated output from downstream pipelines.
- Optimised GenAI API integrations with token management, dynamic prompt truncation, and response caching, significantly cutting LLM latency and per-request API cost for bulk content generation.
- Applied NLP and LLM techniques to classify, process, and optimise content pipelines, improving processing efficiency by ~25% and reducing manual intervention.
- Built and productionised REST APIs in Python/Flask for secure model serving and real-time execution of automation tasks across multiple client websites.
- Deployed and maintained the platform on AWS (EC2, Lambda, S3, RDS); managed SageMaker pipelines that retrain models on schedule as input data drifts.
- Instrumented MongoDB-backed logging of every QA result and automated run, providing full traceability for client audits and debugging model regressions.
- Integrated AWS Lex, Rekognition, and Polly into the platform; configured IAM roles and encryption to maintain security and compliance.

Data Scientist | *Young Minds Technology Solutions Pvt. Ltd., Hyderabad* Sep 2021 – Aug 2023

- Built predictive and deep learning models for business forecasting end-to-end, from raw data through to production deployment.

- Extended into computer vision with OpenCV to extract structured data from images — first production work beyond tabular ML.
- Built Flask and Django applications enabling business teams to consume model outputs directly, removing analyst bottlenecks.
- Maintained and optimised SQL databases feeding Python pipelines, materially improving downstream analytics reliability.

Data Science Intern | *Innomatics Research Labs, Hyderabad*

Jan 2021 – Apr 2021

- Performed EDA and feature engineering on large financial datasets, preparing them for production-grade modelling.
- Built and iteratively refined loan-decision models until precision reached a client-presentable standard.

HR Associate | *Tenpath Solutions, Hyderabad*

Aug 2018 – Nov 2020

- Managed end-to-end recruitment for US and Canadian clients across technical and non-technical roles, building a foundation in stakeholder communication and client management.
- Handled US client accounts, coordinating hiring requirements, screening candidates, and liaising between hiring managers and applicants to close positions efficiently.
- Developed strong analytical and organisational skills that later transferred directly to data pipeline design and project management in ML roles.

KEY PROJECTS

- **Agentic QA Automation Platform** — Python, Flask, LangChain, OpenAI, AWS (EC2, Lambda, S3, RDS), MongoDB
Architected and built a full-stack agentic AI platform that automates website QA at scale. LLM-driven agents orchestrate multi-step validation workflows — invoking SEO checkers, broken-link scanners, content-quality validators, and accessibility tools via function calling. A meta-model layer aggregates per-page scores into a site-level quality index and routes anomalies to human reviewers. Rule-based fallback logic ensures deterministic results when LLM confidence is low. Deployed on AWS with SageMaker-based model retraining, MongoDB audit logging, and Flask REST APIs serving real-time results to client dashboards.
- **Document Intelligence Chatbot (RAG)** — LangChain, Pinecone, Python, OpenAI Embeddings
Built a document intelligence chatbot over enterprise PDFs, reducing manual search time by 80%+. Designed a RAG architecture on Pinecone with hybrid search (Vector + BM25), RecursiveCharacterTextSplitter chunking, and prompt engineering for optimised context handling.
- **COVID-19 Detection from Chest X-Rays** — Python, TensorFlow, Keras (VGG16 / VGG19 / ResNet50)
Trained and benchmarked three CNN architectures on doctor-screened X-ray data, quantifying how preprocessing choices affected classification performance.
- **Sentiment Analysis on Movie Reviews** — Python, RNN, NLP
Built an RNN classifier that handled negation and mid-sentence context shifts noticeably better than classical ML baselines.
- **SVM Model Deployment** — Scikit-learn, Streamlit, Heroku
Took an SVM model from notebook to live web application — first end-to-end deployment experience.
- **Body Posture Controller** — OpenCV, MediaPipe, Tkinter
Used real-time body-landmark tracking to translate posture into character movement controls.
- **CO2 Emission Forecasting** — Python, Statsmodels, ARIMA
Forecast monthly CO2 emissions with Seasonal ARIMA, validating stationarity and seasonal decomposition throughout.

EDUCATION

- **PG Diploma in Data Science (Prodegree)** — Imarticus Learning
- **B.Tech, Electrical and Electronics Engineering** — DMS Sri Venkateswara Hindu College of Engineering, 2018

CERTIFICATIONS & ACHIEVEMENTS

- IBM Certified Specialist — IBM SPSS Modeller Level 1
- Ranked 473rd of 10,000+ participants — Analytics Vidhya Hackathon
- Paper Presentation — Sri Vasavi College of Engineering Fest